

Table 2: Spectral structure of weight updates (FP32 training)
AdamW+GRPO concentrates 90% of update energy in 190/1536 directions (stable rank ≈ 5)

Optimizer	Objective	% Params Changed	$\ \Delta W\ _F$	Rank @90%	σ_k @90%	$\Sigma \sigma_i$ @90%	Stable Rank
AdamW	GRPO	100.0%	0.0327	190 / 1536	6.92e-04	0.332	5.4
Muon	GRPO	99.9%	0.0214	752 / 1536	4.72e-04	0.620	330.1
AdamW	SFT	100.0%	0.0387	450 / 1536	7.28e-04	0.740	10.6
Muon	SFT	99.9%	0.0333	794 / 1536	7.37e-04	0.986	302.9